

Designing a Secure Network Topology (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 03

Student:

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Time on Task:

21 hours, 4 minutes

Progress:

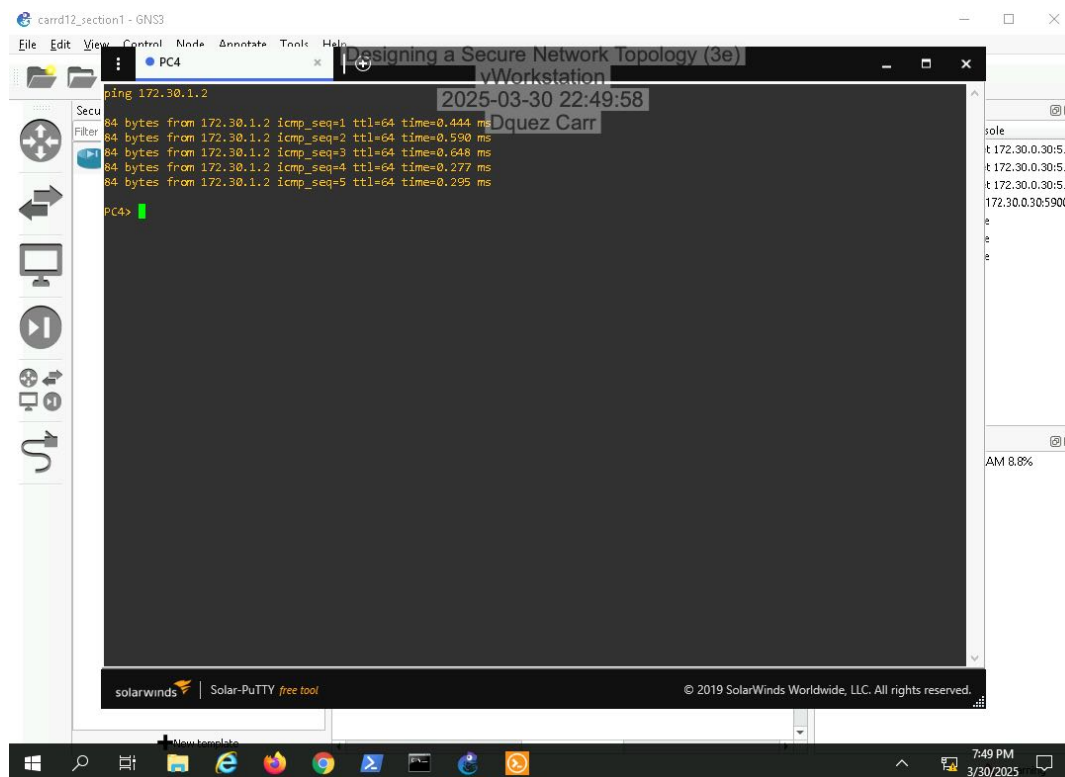
88%

Report Generated: Monday, March 31, 2025 at 2:41 PM

Section 1: Hands-On Demonstration

Part 1: Design a Simple Network Topology

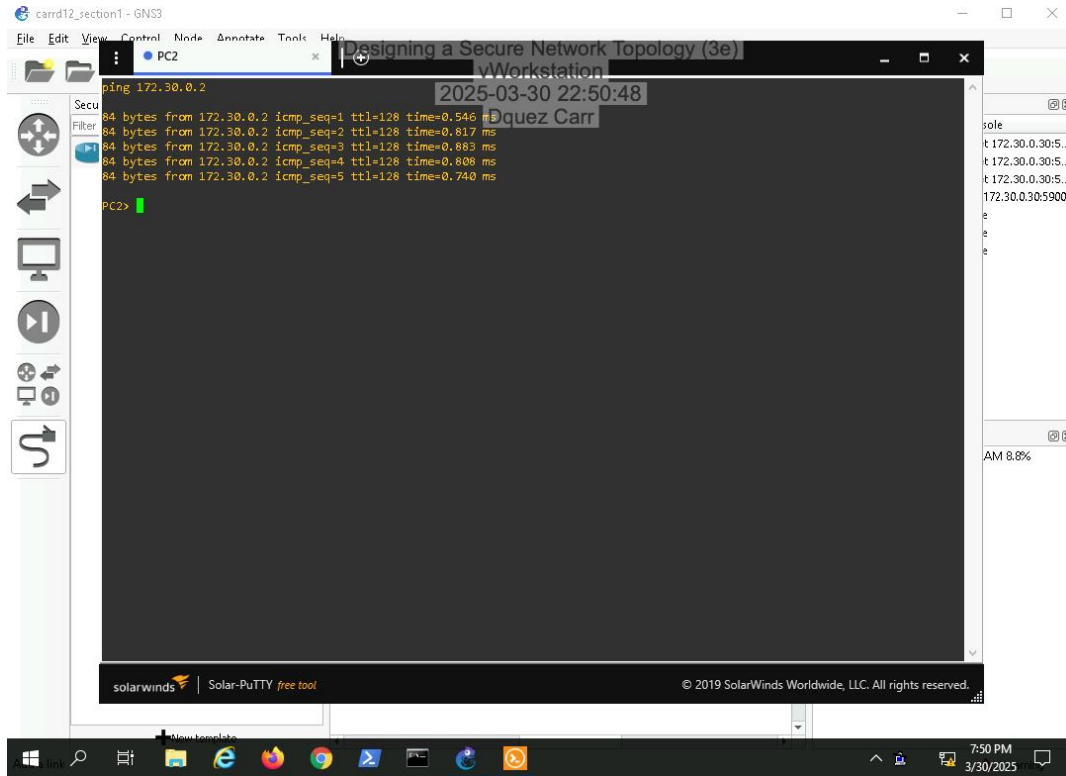
34. Make a screen capture showing the results of the ping attempt from PC4 to PC3.



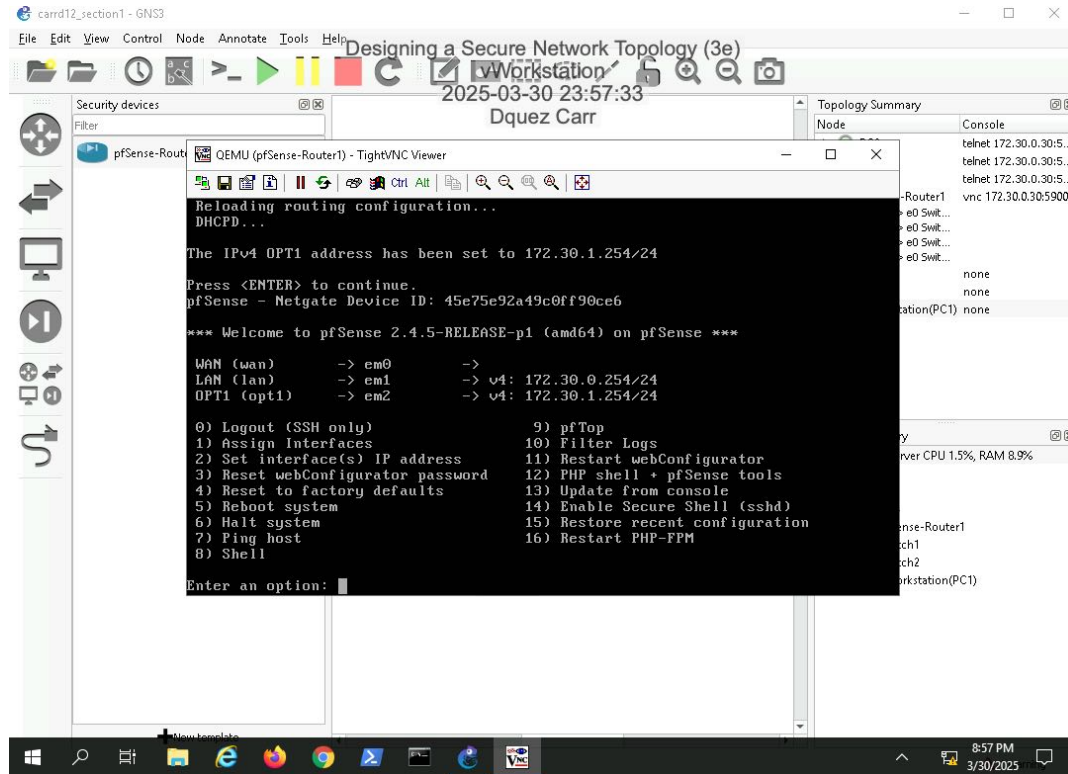
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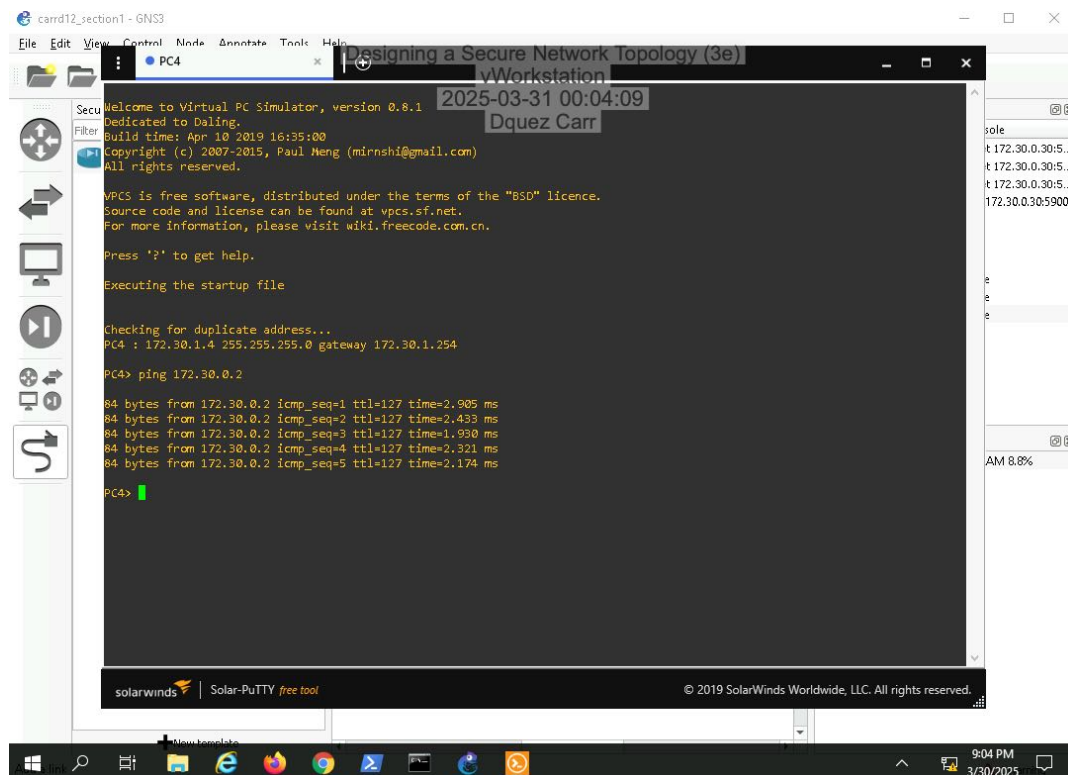
37. Make a screen capture showing the results of the ping attempt from PC2 to the vWorkstation (172.30.0.2).



49. Make a screen capture showing the IP address assignments in the pfSense console.

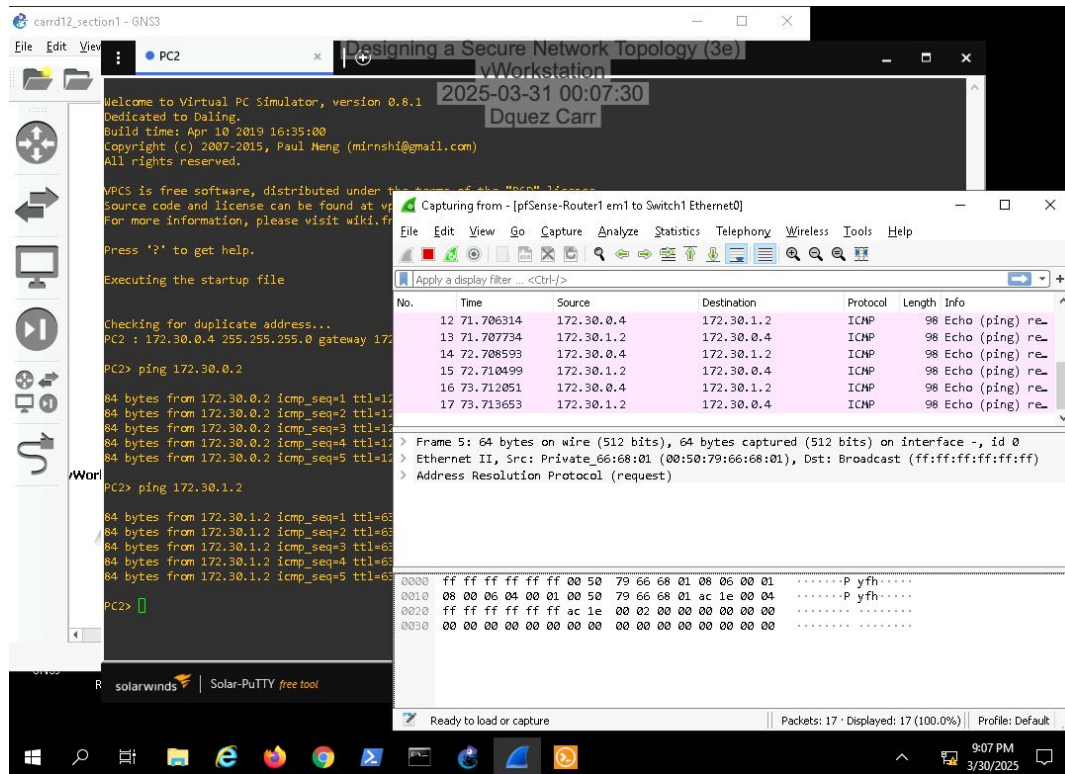


72. Make a screen capture showing the successful ping from PC4 to the vWorkstation.



Part 2: Capture Network Traffic to Validate Connectivity

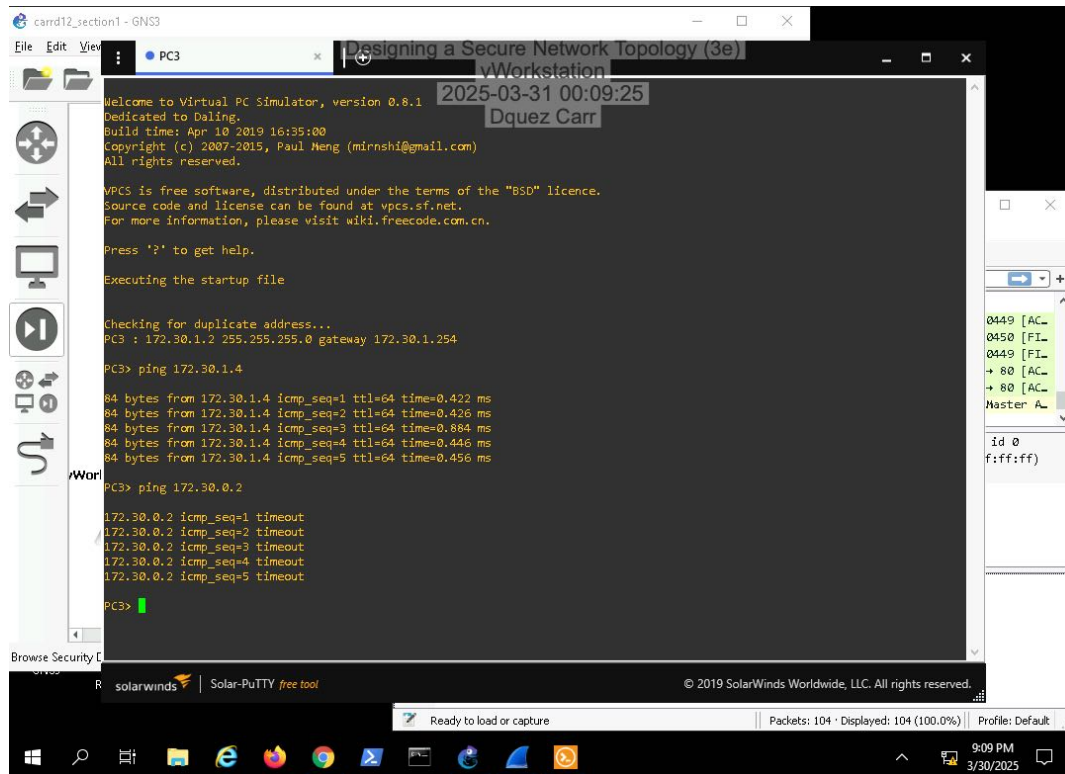
7. Make a screen capture showing the Wireshark capture for the PC2 to PC3 ping request.



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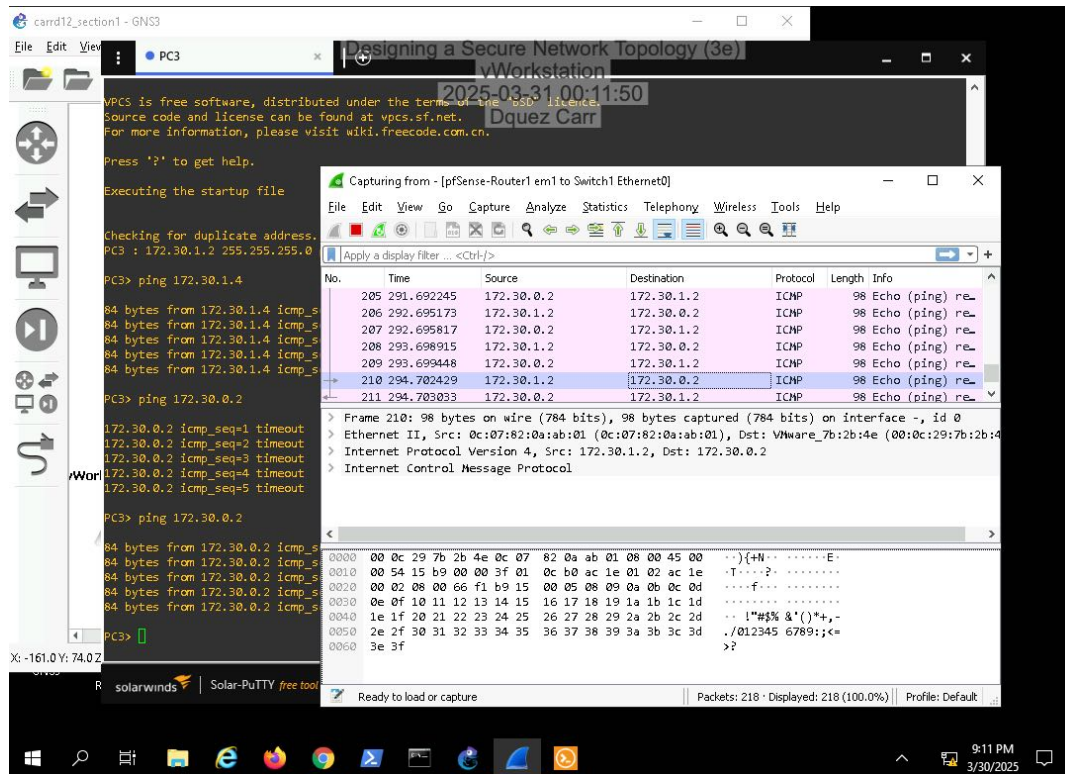
17. Make a screen capture showing the timed out ping request.



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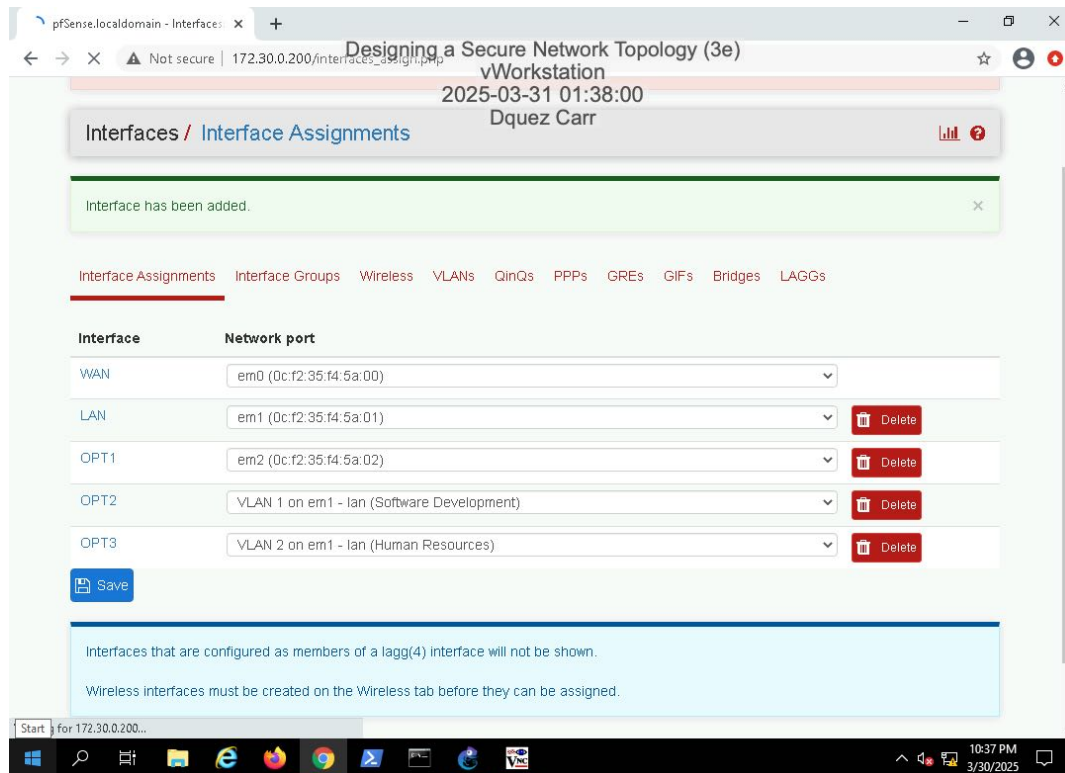
20. Make a screen capture showing the Wireshark capture for the PC3 to vWorkstation ping request.



Section 2: Applied Learning

Part 1: Design a More Complex Network Topology

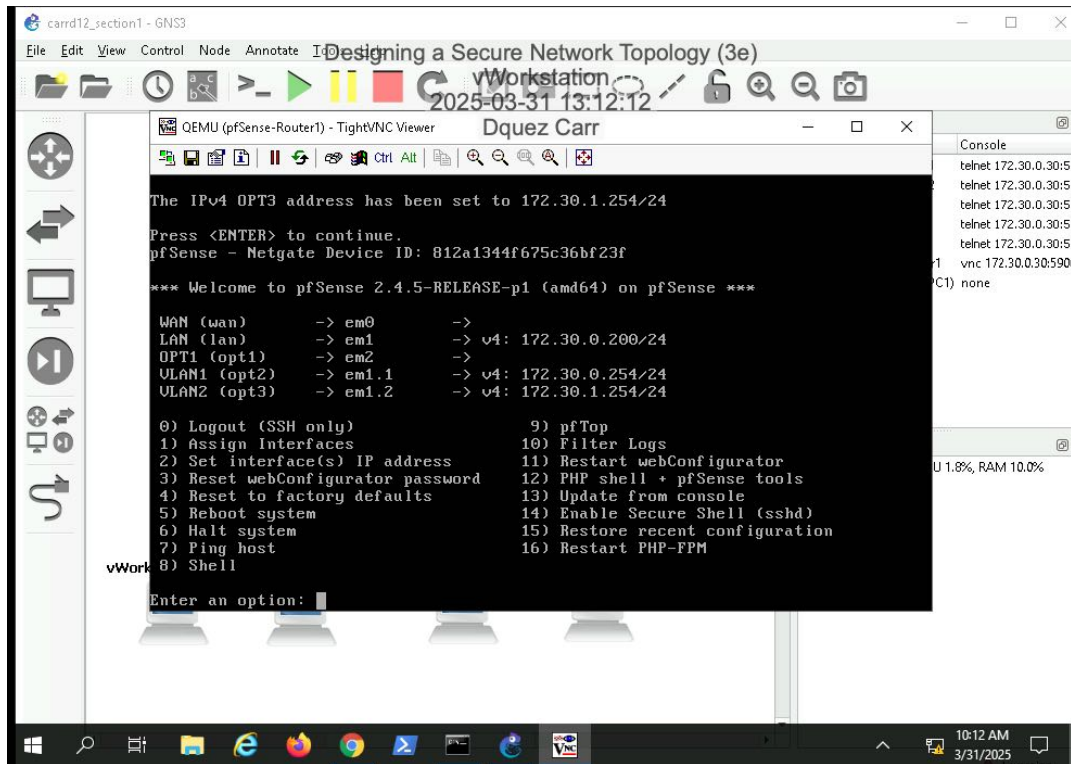
14. Make a screen capture showing the **Interface Assignments** page.



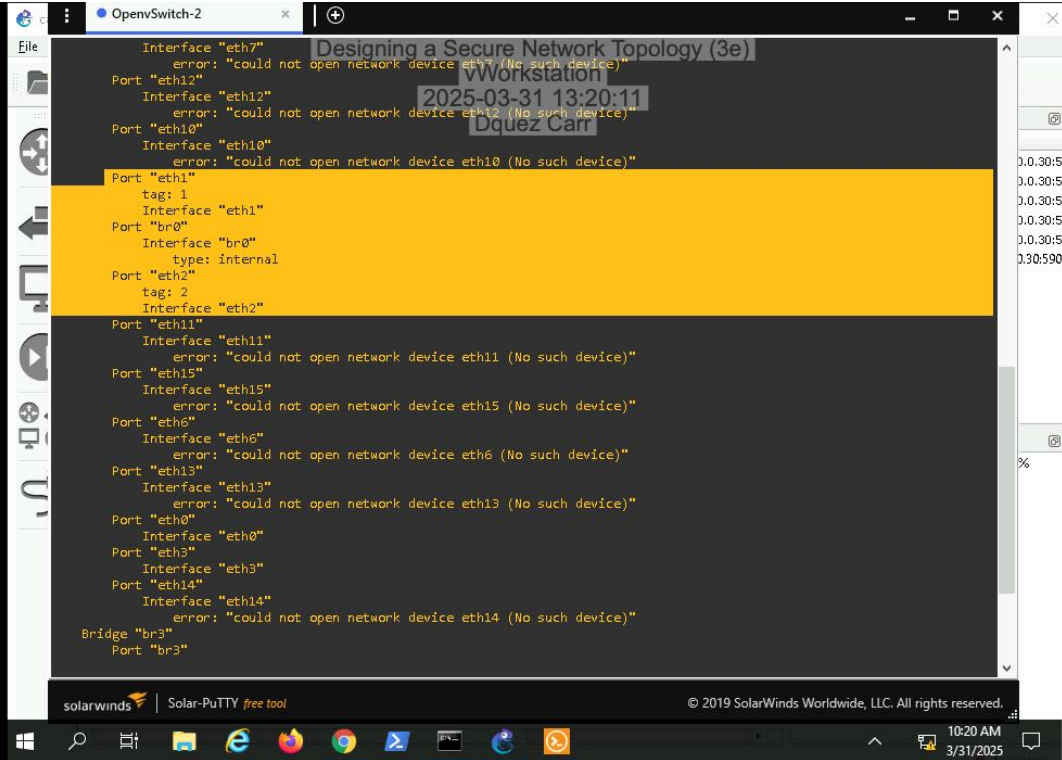
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28. Make a screen capture showing the **updated pfSense router settings with the VLAN IP address assignments.**



36. Make a screen capture showing the **eth1** and **eth2** interfaces with their assigned VLAN tags on **Switch2**.

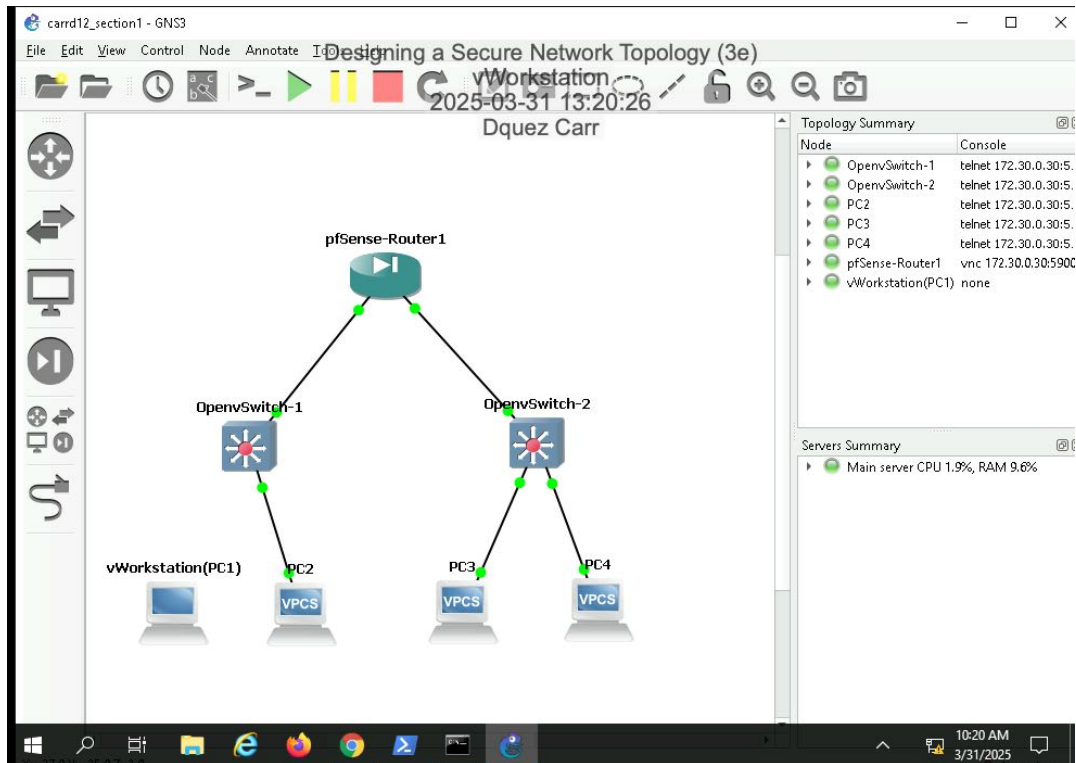


```
Interface "eth7"  
error: "could not open network device eth7 (No such device)"  
Port "eth12"  
Interface "eth12"  
error: "could not open network device eth12 (No such device)"  
Port "eth10"  
Interface "eth10"  
error: "could not open network device eth10 (No such device)"  
Port "eth1"  
tag: 1  
Interface "eth1"  
Port "br0"  
Interface "br0"  
type: internal  
Port "eth2"  
tag: 2  
Interface "eth2"  
Port "eth11"  
Interface "eth11"  
error: "could not open network device eth11 (No such device)"  
Port "eth15"  
Interface "eth15"  
error: "could not open network device eth15 (No such device)"  
Port "eth6"  
Interface "eth6"  
error: "could not open network device eth6 (No such device)"  
Port "eth13"  
Interface "eth13"  
error: "could not open network device eth13 (No such device)"  
Port "eth0"  
Interface "eth0"  
Port "eth3"  
Interface "eth3"  
Port "eth14"  
Interface "eth14"  
error: "could not open network device eth14 (No such device)"  
Bridge "br3"  
Port "br3"
```

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39. Make a screen capture showing the **completed topology**.

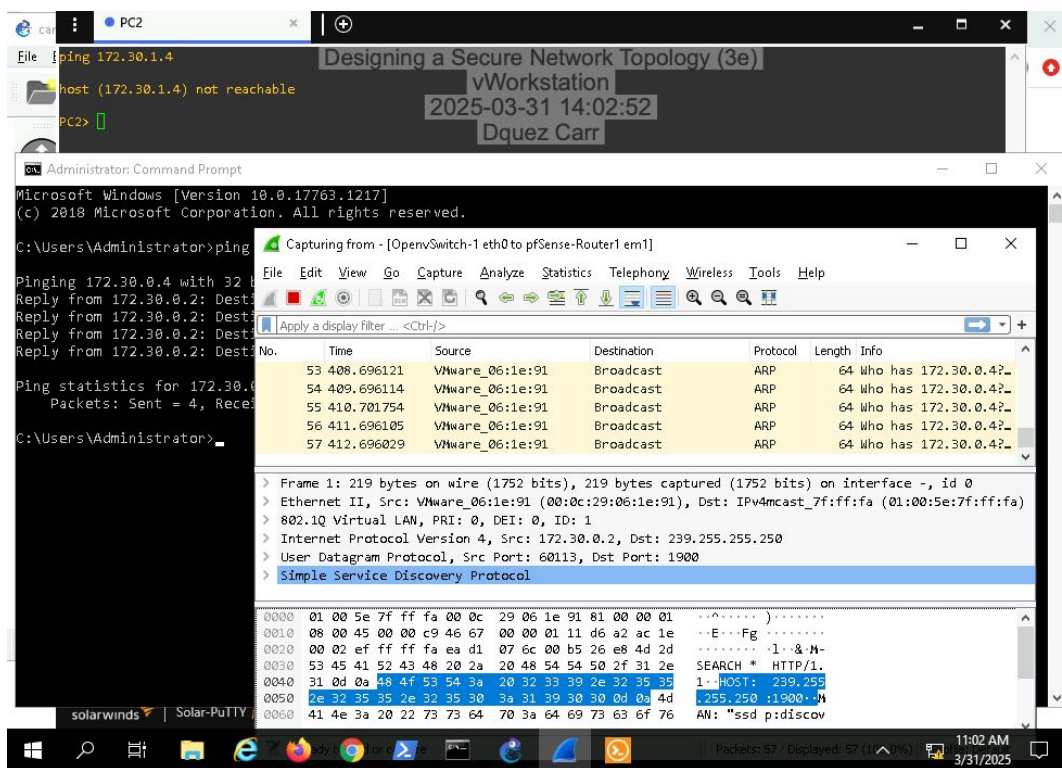


Part 2: Capture Network Traffic to Validate Connectivity

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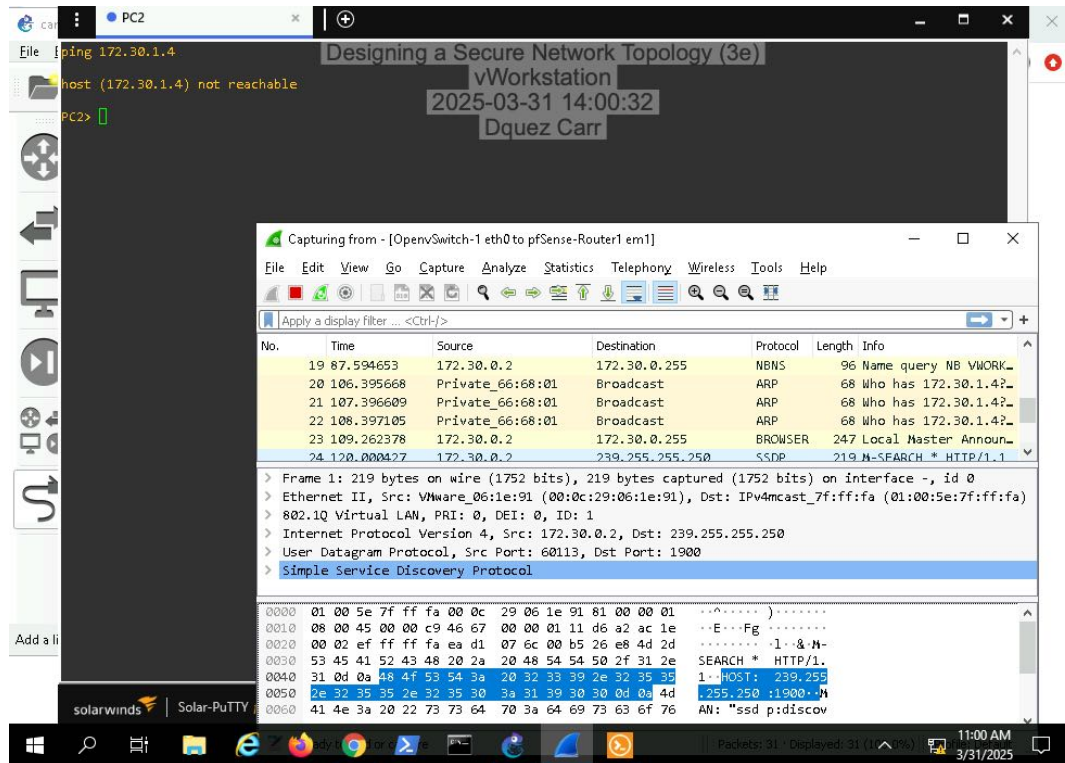
3. Make a screen capture showing the Wireshark capture for the vWorkstation to PC3 ping request.



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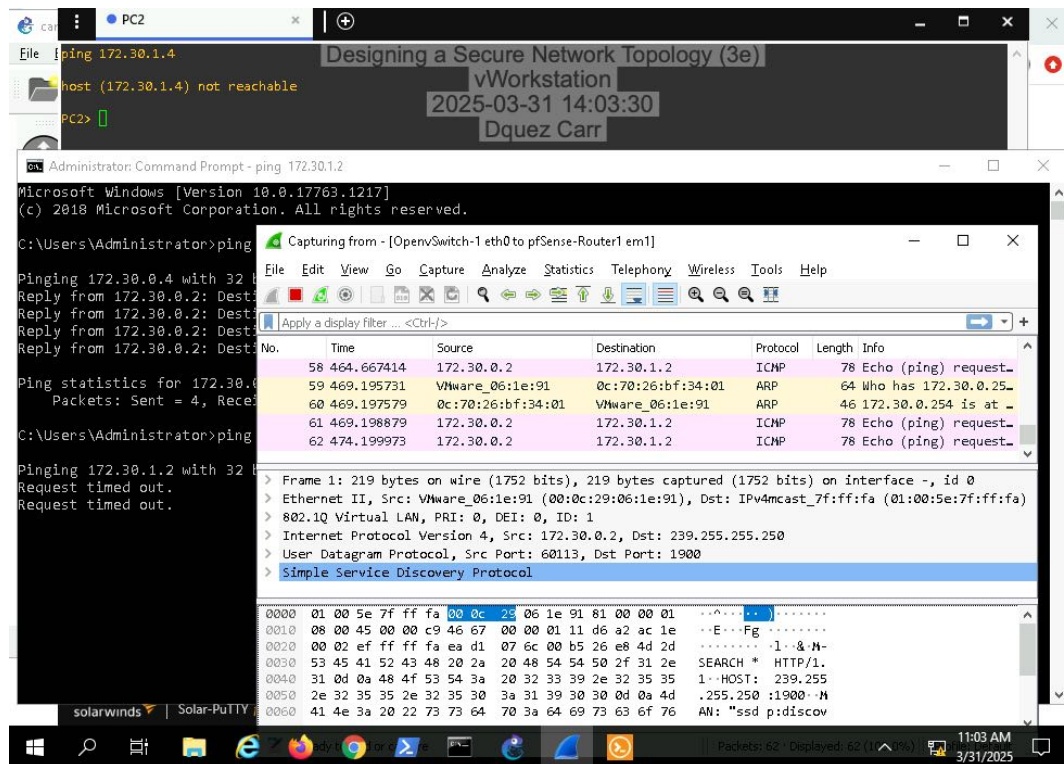
5. Make a screen capture showing the Wireshark capture for the PC2 to PC4 ping request.



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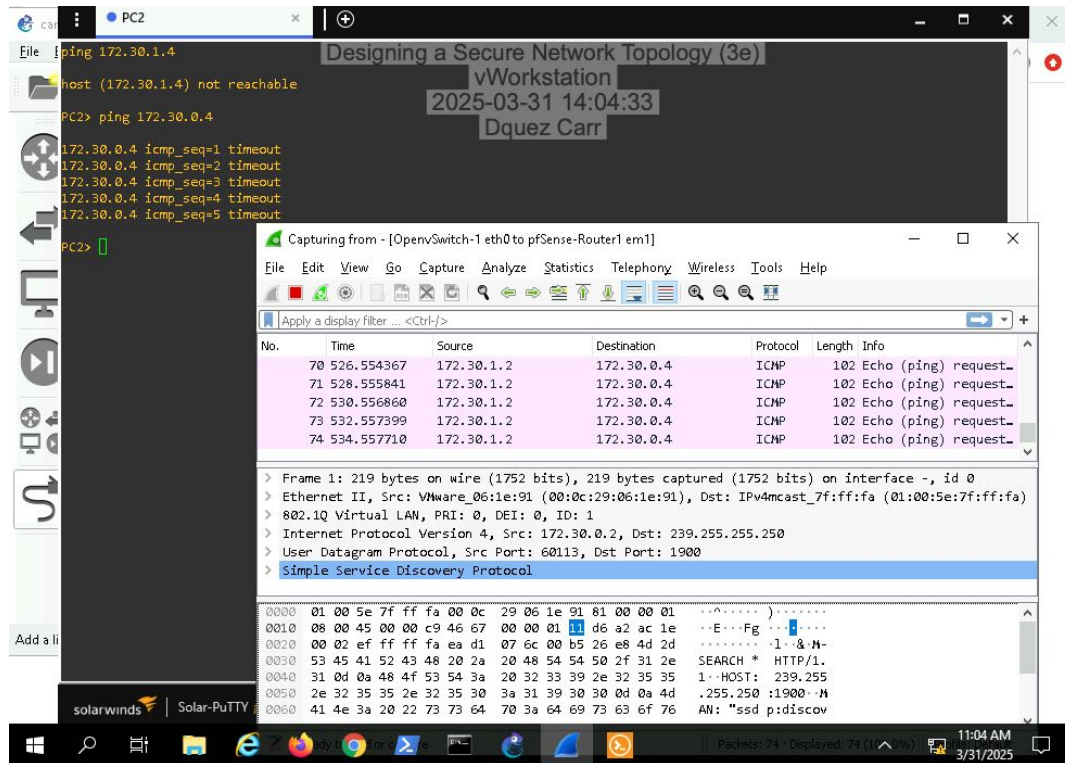
7. Make a screen capture showing the Wireshark capture for the vWorkstation to PC2 ping request.



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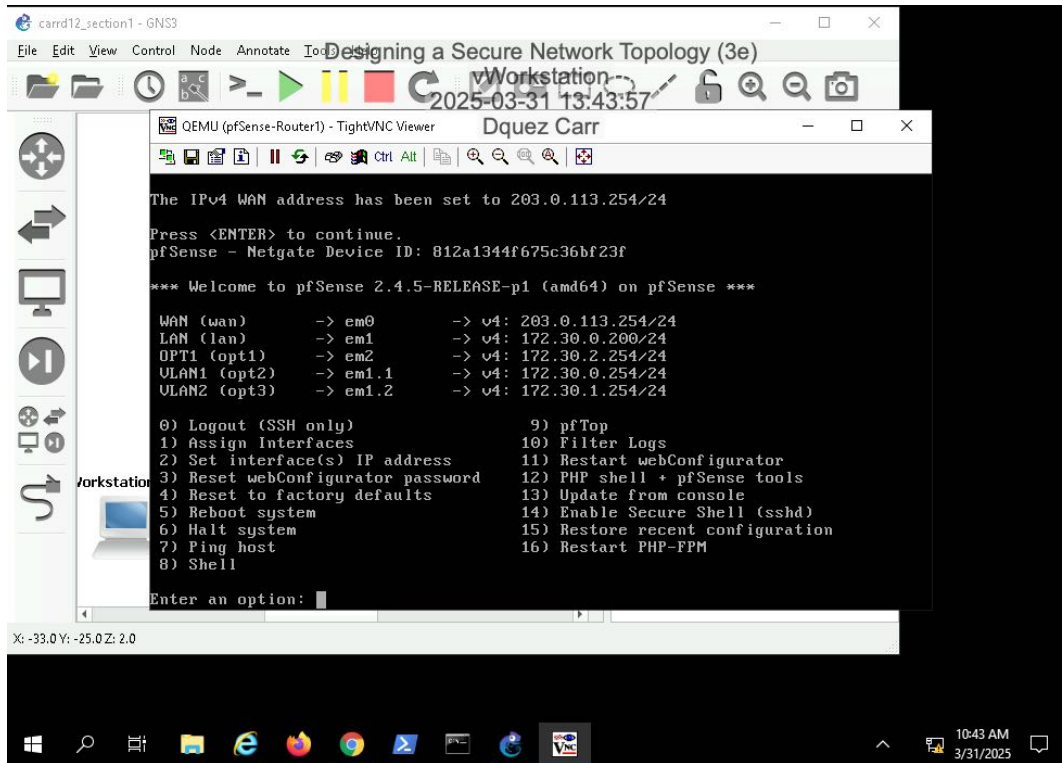
9. Make a screen capture showing the Wireshark capture for the PC2 to PC3 ping request.



Section 3: Challenge and Analysis

Part 1: Enhance the Network Topology with a DMZ

Make a screen capture showing the interface configurations in the pfSense console.



Make a screen capture showing the firewall rule on the WAN interface in the pfSense webConfigurator.

Incomplete

Part 2: Validate DMZ Connectivity

Make a screen capture showing the results of both pings.

Incomplete